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Dear Professor Marc & Selection Team,

I am writing to apply for the EngD position on the interactive reverse logistics map for circular construction (CIRCOLOGIC). This is a topic I really care about. In construction we are already good at taking buildings apart, but a lot of good material still ends up as waste. The main problem is information. People often do not know which materials are available, in what condition, where they are, or when they will need them. So they choose new materials instead of reused ones. A map that shows what is available, a way to compare options on cost, CO_2 and circularity, and a simple decision view for different stakeholders would make reuse a much easier choice. That is the kind of solution I would like to help build.

My master's thesis, **Agile Project Management in the Construction Industry**, looked at how Agile can improve collaboration, continuous risk management, adaptive decision-making and resource efficiency on construction projects. The main outcome was a clearer way to plan and manage work so teams can respond faster to change and use resources better. Completing that work also showed me a limit: better delivery and less waste during construction only cover one stage of sustainability. That raised a bigger question for me: what happens to materials when buildings reach the end of their service life?

While preparing this application, I read some of Professor Voordijk's work on recovering building elements for reuse. His point that reuse needs coordination from the donor building to the target building, not only technical innovation, matched the direction my thesis had already taken me. It also explained why reverse logistics is so hard in practice. CIRCOLOGIC works in that same kind of setting: uncertain, unpredictable and volatile conditions around material availability, quality, timing and cost. Those are the kinds of circumstances I addressed in my thesis, where Agile or hybrid ways of working help teams adapt. CIRCOLOGIC is the right next step for me: it offsets my project and stakeholder experience with the chance to build a digital solution where material information and decisions can meet in one place.

My recent jobs also prepare me for this project. At Aug.e I worked on digital twins for smart buildings. We used a digital environment to understand and improve how a real building performs. For CIRCOLOGIC I see the same idea, but applied to materials: where they are, what can be reused, who is involved, and how they move through reverse logistics. At Hive CPQ I managed large manufacturer datasets and turned complex product information into simple digital solutions that non-technical users could work with. At iostring I worked on a circular construction hypothesis, looking at how Agile project management can bring circular practices into real delivery. Together with my civil and environmental engineering background and my GIS skills, this gives me a base I can bring to the project from day one.

For the last while I have taken a break from work to take care of my newborns. During this time I kept learning on my own about circular construction, reverse logistics and GIS. My background is practical, from real projects and industry roles. What I want next is to take that further in a focused way: reverse-logistics modelling, using GIS to match donor and target materials, and building the CIRCOLOGIC solution with partners like Dura Vermeer and TNO. I am ready to relocate to the Netherlands for this EngD.

I would really like to do this at the University of Twente. I like that the EngD combines advanced study with a design project in industry. I want to work in a design-science way: define real use cases with partners, test the solution in a (de)construction pilot, and improve it with the people who will use it.

I would be glad to contribute to CIRCOLOGIC and to grow as an engineer who builds solutions that people actually use. I would be happy to talk more about how I can fit into the project. Thank you for reading my application.

Yours sincerely,
Pooja Awasare